SWINBURNE VIETNAM

HO CHI MINH CAMPUS

COS30018 - COS30018-Intelligent Systems

Week 5: Report

INSTRUCTOR: MR. KY TRUNG PHAM

STUDENT NAME: LÊ TIẾN DŨNG-SWS00683

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1. Beyond just the Close price, how do features like Volume or High/Low provide extra context to a Deep Learning model?

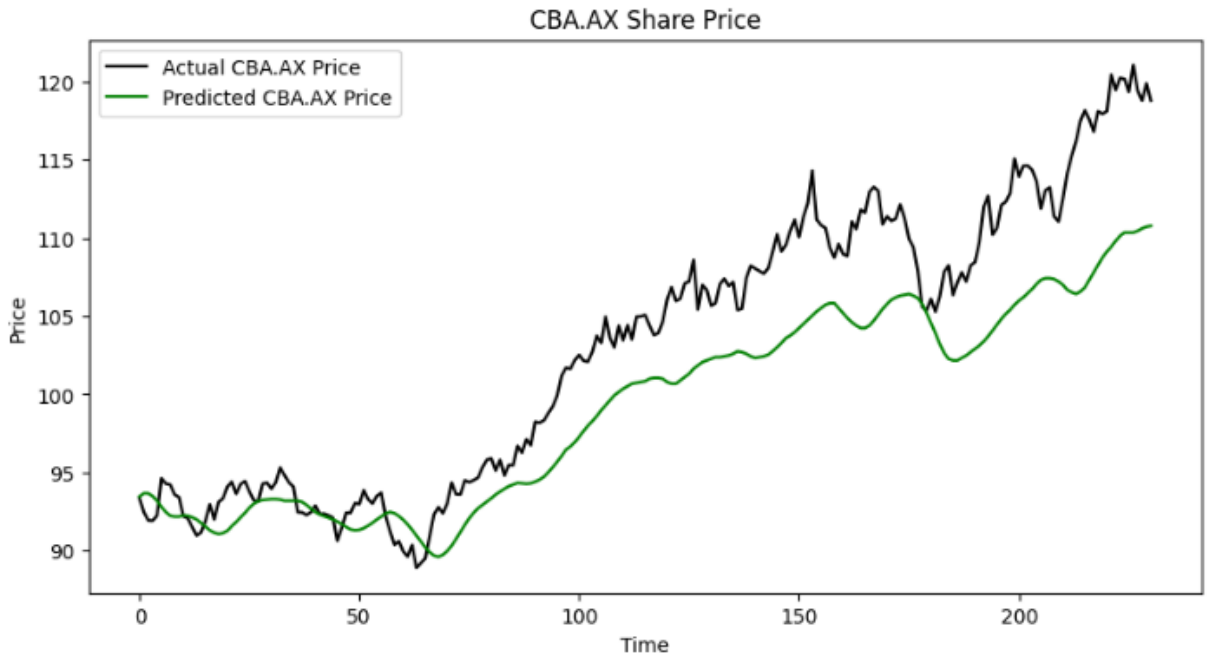
Using only the closing price gives the model a limited view of the market. Adding High, Low, and Volume gives more context about what happened during each trading day. High and Low show the price range and help the model understand volatility. Volume shows how much trading activity happened, which may indicate stronger market interest or pressure. Because of this, the model can learn patterns from both price movement and trading behaviour, instead of relying on just one value.

2. How does your system handle NaN values in the historical stock data? Why is it better to handle these during preprocessing rather than letting the model process them?

The system handles NaN values during preprocessing using forward fill and backward fill. This means missing values are replaced using nearby valid values before the data is scaled or turned into sequences. This is better than leaving NaN values for the model because neural networks do not work well with missing data. If NaN values are left in the dataset, they can break scaling, cause unstable training, or lead to invalid predictions. Cleaning them early makes the input data consistent and usable.

3. Explain the logic of scaling your data using MinMaxScaler. Why must we store the scaler object to inverse transform the results later?

MinMaxScaler is used to transform the values into a similar range, usually between 0 and 1. This is important because the features in stock data can be very different in size. For example, Volume can be much larger than Close, High, or Low. If the data is not scaled, the larger values may dominate the training process and make learning less stable. The scaler object must be stored because the model predicts values in scaled form. To convert those predictions back into real stock prices, the same fitted scaler must be used again through inverse transformation.

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1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 48ms/step
Predicted next closing price: 110.87780144065619
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